Experiment no:6

Determination and plotting of Capacitance of parallel plate capacitor in three different Dielectric medium with specific area, dielectric thickness and various dielectric constant.

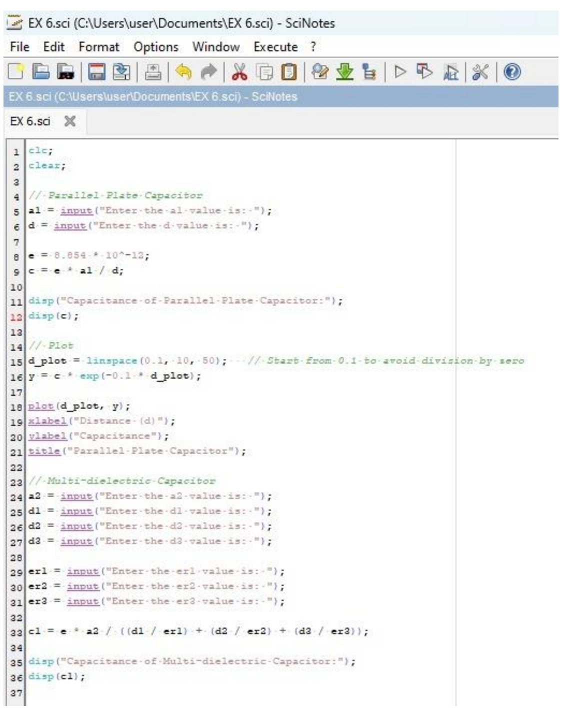
Aim:

To write program for capacitance of parallel plate capacitor in three different dielectric media using SCILAB.

Software required:

•SCILAB version 6.0.1

Program:

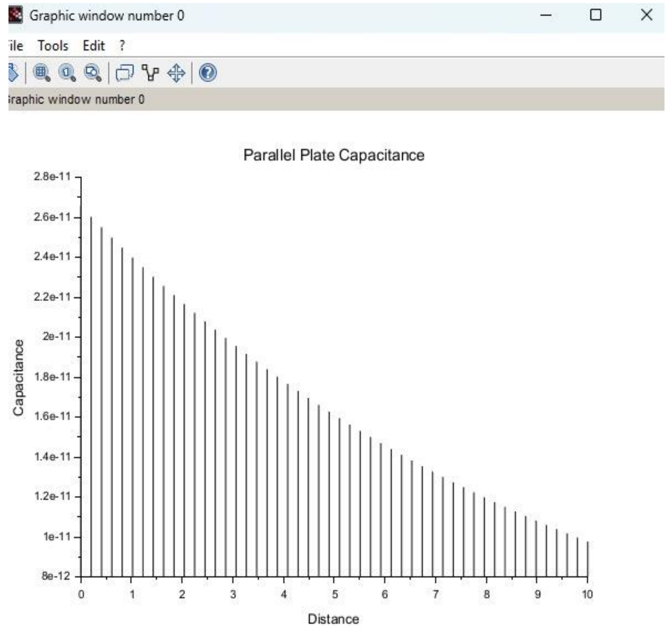


EX 6.sci (C:\Users\user\Documents\EX 6.sci) - SciNotes

```
clc;
clear;

// Parallel Plate Capacitor
a1 = input("Enter the a1 value is: ");
d = input("Enter the d value is: ");

e = 8.854 * 10^-12;
c = e * a1 / d;

disp("Capacitance of Parallel Plate Capacitor:");
disp(c);

// Plot
d_plot = linspace(0.1, 10, 50);   // Start from 0.1 to avoid division by zero
y = c * exp(-0.1 * d_plot);

plot(d_plot, y);
xlabel("Distance (d)");
ylabel("Capacitance");
title("Parallel Plate Capacitor");

// Multi-dielectric Capacitor
a2 = input("Enter the a2 value is: ");
d1 = input("Enter the d1 value is: ");
d2 = input("Enter the d2 value is: ");
d3 = input("Enter the d3 value is: ");

er1 = input("Enter the er1 value is: ");
er2 = input("Enter the er2 value is: ");
er3 = input("Enter the er3 value is: ");

c1 = e * a2 / ((d1 / er1) + (d2 / er2) + (d3 / er3));

disp("Capacitance of Multi-dielectric Capacitor:");
disp(c1);
```

Output:

```
Enter the a1 value is: 0.0003

Enter the d value is: 0.001

  "Capacitance of Parallel Plate Capacitor:"
   2.656D-12
Enter the a2 value is: 0.0006

Enter the d1 value is: 0.002

Enter the d2 value is: 0.003

Enter the d3 value is: 0.004

Enter the er1 value is: 4

Enter the er2 value is: 6

Enter the er3 value is: 8

  "Capacitance of Multi-dielectric Capacitor:"
   3.542D-12
```



Theoretical calculation :

EXP NO : 06

Determination and plotting of capacitance of parallel plate capacitor in three different dielectric medium with specific area, dielectric thickness and various dielectric constant

Output :

Enter the a1 value is : 3 cm^2
Enter the d value is : 1 mm
Enter the a2 value is : 6 cm^2
Enter the d1 value is : 2 mm
Enter the d2 value is : 3 mm
Enter the d3 value is : 4 mm
Enter the er1 value is : 4
Enter the er2 value is : 6
ε_{r3} : 8

Practical output:

"Capacitance of parallel plate Capacitor :"
2.656D −12

"Capacitance of Multi - dielectric capacitor:"
3.542D - 12.

Theoretical calculation :

$\varepsilon_0 = 8.854 \times 10^{-12}$ F/m

$C = \frac{\varepsilon_0 A}{d}$ $\quad A = 3\,cm^2 = 3 \times 10^{-4}\,m^2$

$d = 1\,mm = 1 \times 10^{-3}\,m$

$C = \frac{8.854 \times 10^{-12} \times 3 \times 10^{-4}}{1 \times 10^{-3}}$

$C = 2.6562 \times 10^{-12}\,F$

$C = 2.6562 \times 10^{-12}\,F$

$C = 2.6562\,pF$

Capacitance of parallel plate capacitor

$C = 2.6562\,pF$

Capacitance of each dielectric

Dielectric 1 $\quad \varepsilon_{r1} = 4$

$d_1 = 2\,mm$

$C_1 = \frac{8.854 \times 10^{-12} \times 4 \times 2 \times 10^{-4}}{2 \times 10^{-3}}$

$C_1 = 5.3124 \times 10^{-12}\,F$

$C_1 = 5.3124\,pF$

Dielectric 2 $\quad \varepsilon_r = 6,\ d_2 = 3mm$

$C_2 = \frac{8.854 \times 10^{-12} \times 6 \times 3 \times 10^{-4}}{3 \times 10^{-4}}$

$C_2 = 5.3124 \times 10^{-12}\,F$

$C_2 = 5.3124\,pF$

Dielectric 3 $\quad \varepsilon_r = 8,\ d_3 = 4mm$

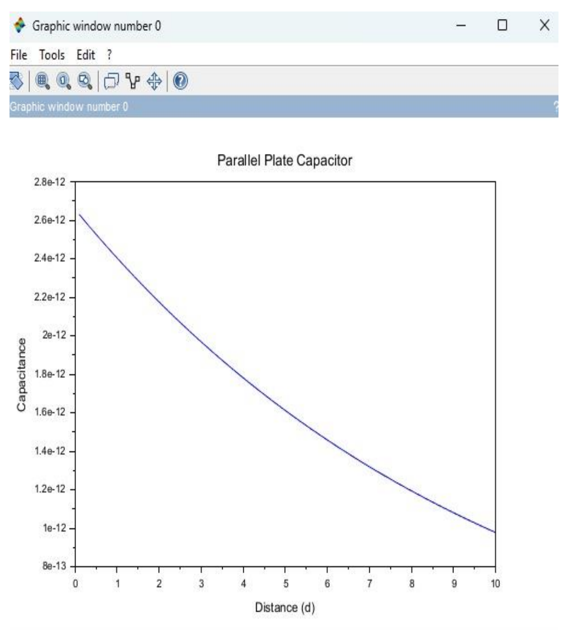
$C_3 = \frac{8.854 \times 10^{-12} \times 8 \times 9 \times 10^{-4}}{4 \times 10^{-3}} = 5.3124 \times 10^{-12}\,F = 5.3124\,pF$

Case 1. AIR:

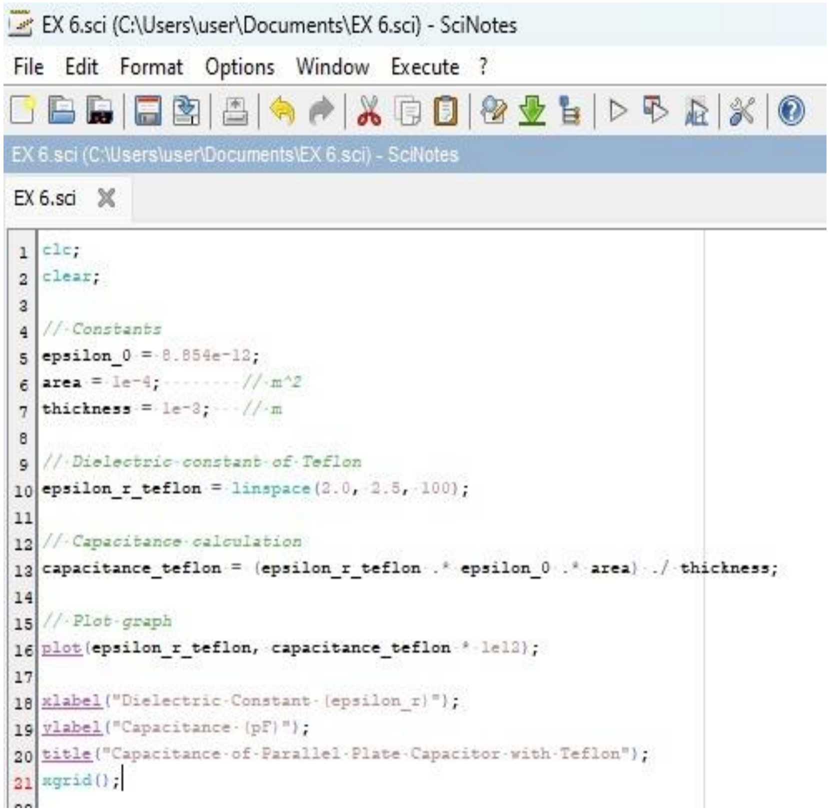
```
// Plot
d_plot = linspace(0.1, 10, 50);   // Start from 0.1 to avoid division by zero
y = c * exp(-0.1 * d_plot);

plot(d_plot, y);
xlabel("Distance (d)");
ylabel("Capacitance");
title("Parallel Plate Capacitor");

// Multi-dielectric Capacitor
a2 = input("Enter the a2 value is: ");
d1 = input("Enter the d1 value is: ");
d2 = input("Enter the d2 value is: ");
d3 = input("Enter the d3 value is: ");

er1 = input("Enter the er1 value is: ");
er2 = input("Enter the er2 value is: ");
er3 = input("Enter the er3 value is: ");

c1 = e * a2 / ((d1 / er1) + (d2 / er2) + (d3 / er3));

disp("Capacitance of Multi-dielectric Capacitor:");
disp(c1);
```

Output:

```
Enter the a1 value is: 0.0003

Enter the d value is: 0.001

  "Capacitance of Parallel Plate Capacitor:"
   2.656D-12
Enter the a2 value is: 0.0006

Enter the d1 value is: 0.002

Enter the d2 value is: 0.003

Enter the d3 value is: 0.004

Enter the er1 value is: 4

Enter the er2 value is: 6

Enter the er3 value is: 8

  "Capacitance of Multi-dielectric Capacitor:"
   3.542D-12
```

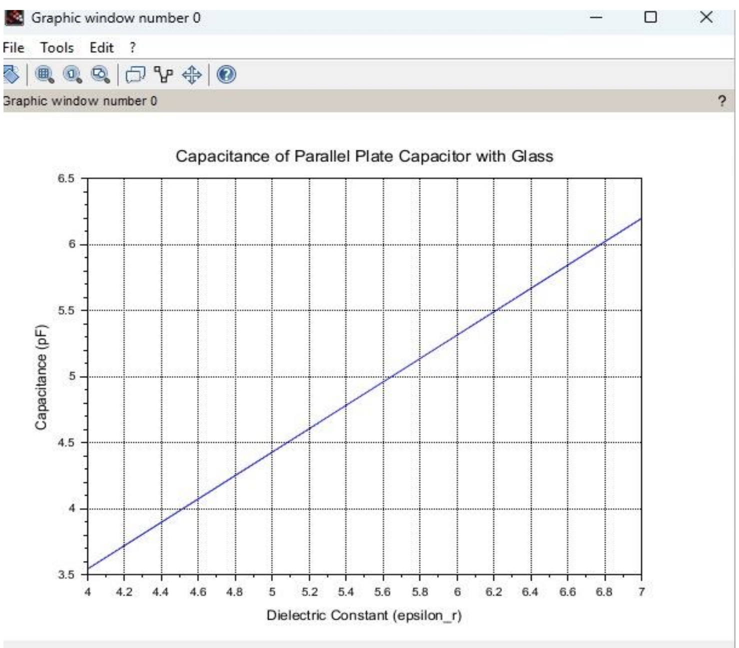


Case 2. Glass:

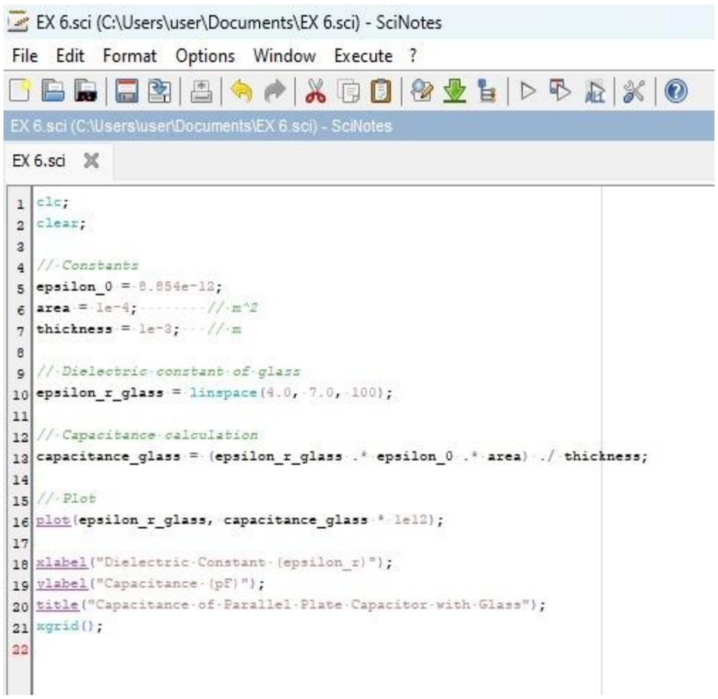



```
clc;
clear;

// Constants
epsilon_0 = 8.854e-12;
area = 1e-4;        // m^2
thickness = 1e-3;   // m

// Dielectric constant of Teflon
epsilon_r_teflon = linspace(2.0, 2.5, 100);

// Capacitance calculation
capacitance_teflon = (epsilon_r_teflon .* epsilon_0 .* area) ./ thickness;

// Plot graph
plot(epsilon_r_teflon, capacitance_teflon * 1e12);

xlabel("Dielectric Constant (epsilon_r)");
ylabel("Capacitance (pF)");
title("Capacitance of Parallel Plate Capacitor with Teflon");
xgrid();
```

Output:

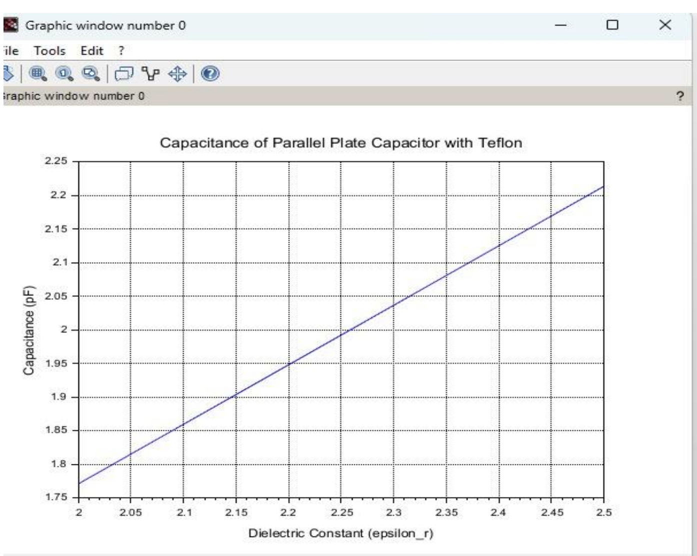


Case 3. Teflon:




```
clc;
clear;

// Constants
epsilon_0 = 8.854e-12;
area = 1e-4;        // m^2
thickness = 1e-3;   // m

// Dielectric constant of glass
epsilon_r_glass = linspace(4.0, 7.0, 100);

// Capacitance calculation
capacitance_glass = (epsilon_r_glass .* epsilon_0 .* area) ./ thickness;

// Plot
plot(epsilon_r_glass, capacitance_glass * 1e12);

xlabel("Dielectric Constant (epsilon_r)");
ylabel("Capacitance (pF)");
title("Capacitance of Parallel Plate Capacitor with Glass");
xgrid();
```

Output:



Result :

Thus the program was verified for capacitance of parallel plate capacitor in three different dielectric media using SCILAB was successfully.